PROFESSIONAL SMART EYEWEAR

Buyer Decision Guide

A practical guide to task, configuration, evidence and commercial decisions.

TASK / WEARER / SYSTEM / EVIDENCE / SCALE

Every performance claim needs a workload, a configuration and a measurement.

Engineering AI and AR eyewear for real work.
September 2026 · www.topaiglasses.com

The task is the first specification.

A professional wearable earns its place when a person can complete a useful task, trust the result and repeat it.

A maintenance technician needs a readable asset number. An inspector needs a record that can be traced to the right job. A remote specialist needs enough context to make the next decision. Each workflow creates a different product brief.

That brief reaches beyond camera resolution or an AI model. It includes the face that wears the device, the environment around it, the systems it connects to and the team that will keep it working.

LENS FOR THE DECISION	WHAT SUCCESS REQUIRES
The worker	Useful output, tolerable fit and heat, accessible interaction and recovery when something fails.
The organization	Appropriate protection, controlled data, compatible software, measurable benefit and supportable service cost.
The product team	A buildable configuration, repeatable verification and a production process that preserves the result.

Use this guide to turn a feature request into a decision and a request for evidence. This is SmartXY editorial planning guidance, not a certification scheme or a standardized laboratory test.

Read: decisions p3 · workflow p4 · concept views p5-7 · engineering p8-19 · delivery p20-24 · checklist p25 · collaboration p26 · sources p27.

THE QUESTION TO BRING TO THE MEETING

Which task becomes easier, and what evidence would convince the person responsible for that task?

Twelve statements. One shared language.

Use these exact statements across the guide, technical reference and launch materials.

C01

The task is the first specification.

C02

Every performance claim needs a workload, a configuration and a measurement.

C03

Put the workload next to the battery-life claim.

C04

Heat sets the terms of sustained use.

C05

Megapixels describe the sensor. Test the task.

C06

A voice feature is an entire exchange.

C07

Fit determines who can use the product.

C08

A protection claim belongs to a configuration.

C09

“SDK available” starts the conversation.

C10

Offline must survive the whole workflow.

C11

The spare lens belongs in the business case.

C12

Each gate should earn the next investment.

THE QUESTION TO BRING TO THE MEETING

For every decision, name an owner, an acceptance criterion and the evidence still missing.

Specify the workflow before the hardware.

Start with a primary task and a baseline. Then decide which functions earn their weight, power and integration cost.

TASK PATTERN	WHAT TO VALIDATE FIRST
Inspection capture	Can the operator obtain the required image, attach it to the correct asset and recover from an incomplete upload?
Label or code reading	Can the system return the exact required field at the actual distance, illumination and motion?
Remote assistance	Can both parties exchange usable information under the available bandwidth, noise and wearing conditions?
In-view guidance	Can the wearer read and act on the information without unacceptable obstruction, distraction or delay?

When a display earns its place

An in-view display can support silent prompts, text or spatial guidance. It also creates optical, fit, power and attention requirements. Evaluate legibility across ambient light, eye position and prescription use, together with the wearer's ability to see the task. Choose display architecture from those requirements.

Write the baseline before the pilot

Record how the task is performed today: completion time, errors, interruptions, repeat visits and documentation quality. Include training and recovery time in the proposed workflow. An attractive demo is useful evidence of a demonstrated function; investment needs evidence of operational benefit.

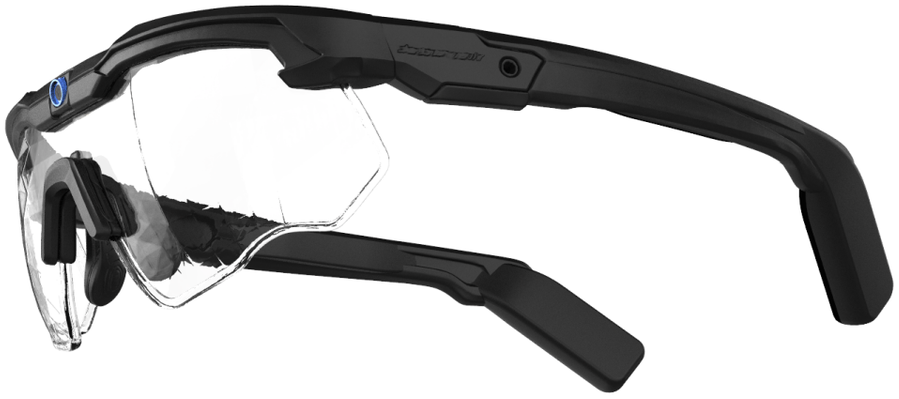
THE QUESTION TO BRING TO THE MEETING

What is the smallest complete workflow that would justify deployment?

CONCEPT DESIGN REFERENCE / 01

A concept for professional smart eyewear.

Overall form: a visual starting point for product definition and engineering discussion.



OVERALL FORM / OBLIQUE VIEW

Review the front lens geometry, nose interface and temple volumes together. Define the imaging path, internal packaging and service interfaces against the intended workflow.

CONCEPT DESIGN REFERENCE / SMARTXY / VANDA

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CONCEPT DESIGN REFERENCE / 02

Explore fit from the side.

Headform side view: discuss temple reach, ear clearance and assembly interfaces.



WEARING REFERENCE / SIDE VIEW

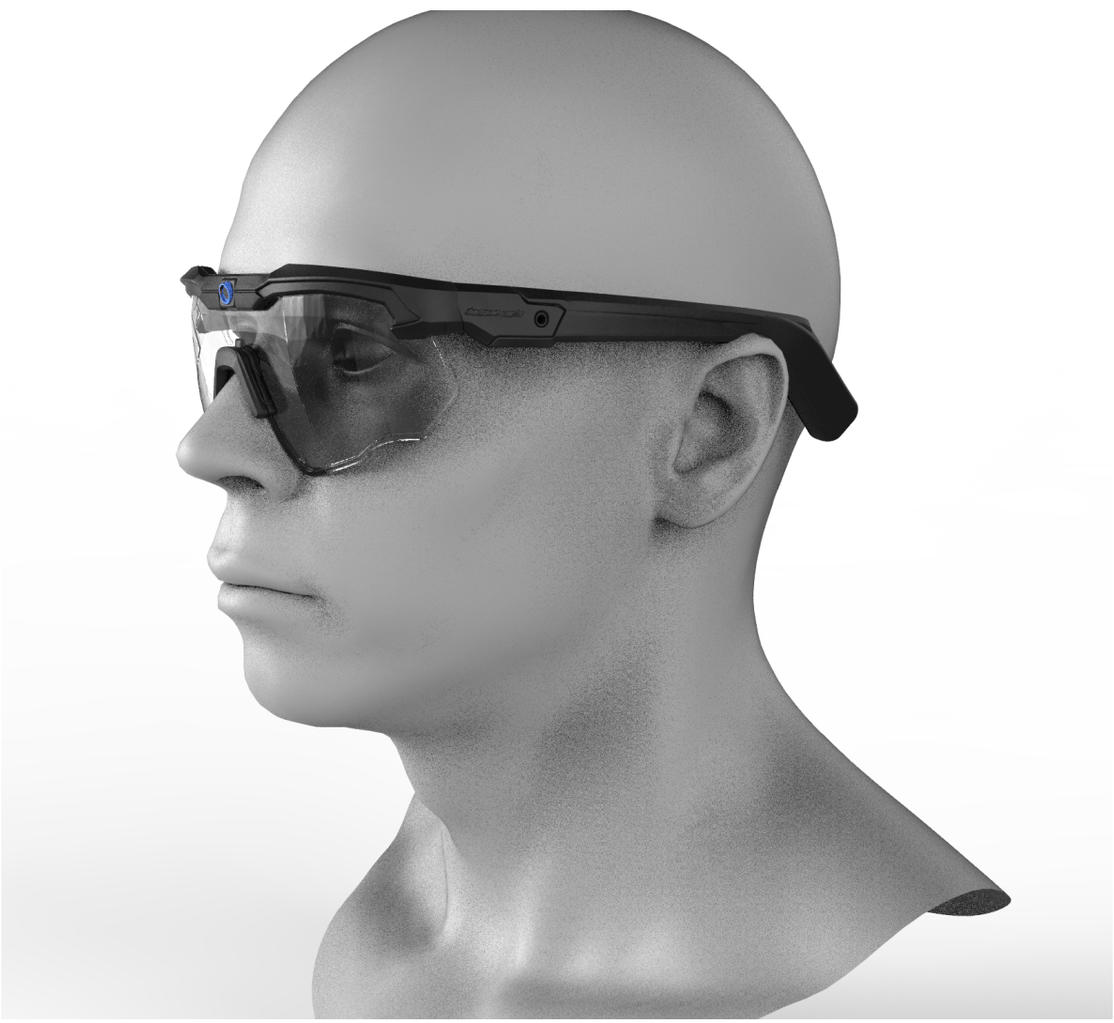
Use physical prototypes to evaluate contact, balance and fit across the intended users. Edited concept view; detached component removed for presentation.

CONCEPT DESIGN REFERENCE / SMARTXY / VANDA

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Review the wearing envelope.

Opposite-side headform view: inspect front coverage, nose contact and the temple path.



WEARING REFERENCE / OPPOSITE SIDE

Evaluate prescription options, eye clearance, stability during movement and compatibility with other equipment. Confirm the wearing envelope through representative fit trials and the selected assessment route.

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A protection claim belongs to a configuration.

Define the hazards and intended use before selecting the assessment route.

MARKET	REFERENCE AND DECISION
United States	ANSI/ISEA Z87.1-2025 has been published. OSHA adoption and the publisher's latest edition are separate questions. Establish the applicable route, protector type and marking evidence. [1,2]
Canada	CSA now lists Z94.3:26 as active; its Z94.3:20 (R2025) listing is withdrawn. Confirm the edition required by the workplace, jurisdiction and assessment route. [3]
European Union	Review EN ISO 16321-1:2022, its A1:2025 amendment and the current harmonized-standard references under the PPE framework. Confirm the applicable transition and conformity route. [4]

Professional eyewear may operate where eye protection is unnecessary, with assessed compatible protection, or as an integrated protector. These are different briefs. Select the arrangement from the workplace assessment and intended claim.

Control the lens, frame, side coverage, camera openings, batteries, prescription parts and accessories. Review retention, added mass, internal components and eye clearance. Impact evidence does not by itself establish suitability for chemical splash, welding, lasers, electrical arc hazards or explosive atmospheres.

A development target, a test report, a declaration and third-party certification describe different evidence. Use the term justified by the completed route. The publication of a standard does not certify a product.

Standards status reviewed 8 September 2026. Exact requirements come from the applicable full standards and assessment route; see p27.

THE QUESTION TO BRING TO THE MEETING

Which exact configuration and intended use does the evidence cover?

Interfaces carry several jobs at once.

A joint may retain a lens, survive deformation, resist ingress and tolerate servicing. Evaluate those jobs together.

INTERFACE	COUPLED QUESTIONS
Lens retention	How does it behave under loading, assembly variation and repeated replacement? Is it actually part of the electronics enclosure?
Acoustic openings	What changes when membranes are wet, dirty, aged or assembled with production variation?
Battery and charging	How are contact wear, corrosion, debris, incorrect insertion and electrical faults controlled?
Housing joints	How do material choice, bond preparation, residual stress and movement affect integrity?

High-impact performance and a high IP rating are not inherently incompatible. Their interaction depends on the enclosure and interface design. IEC 60529 classifies ingress protection; it does not establish resistance to every cleaner, exposure or service condition. Select the IP target from the workflow. [5]

Add credible sequences to the engineering plan

Examples include lens servicing followed by ingress evaluation, or cleaning and thermal ageing followed by mechanical assessment. Agree sequences and acceptance criteria with the responsible specialists. Distinguish supplemental engineering tests from the formal conformity sequence.

Define inspection and retirement after impact. Continued electronic function does not establish that damaged protective equipment remains suitable for use.

THE QUESTION TO BRING TO THE MEETING

Which interface changes after real use, and how will that change be detected?

Choose the geometry for the whole system.

Coverage, optics, integration, fit and service all compete for the same space.

OPTION	POTENTIAL VALUE	WORK TO RESOLVE
Single shield	Continuous front coverage.	Optical quality, retention, ventilation and insert clearance.
Dual lens	Familiar form and prescription supply options.	Side coverage, bridge stiffness, lens retention and electronics packaging.
Over-glasses	Accommodation of an existing prescription pair.	Stable combined fit, visibility, fogging and compatibility with other PPE.

Coverage is three-dimensional

Base curve alone cannot define protection or fit. Evaluate wrap, frame dimensions, lens shape, side geometry and position on the face. Allow space for eyelashes, prescription components and the expected movement of the assembly.

Give the camera an optical specification

A separate window and a view through the protective lens create different imaging and structural questions. Compare focus, distortion, reflections, transmission, contamination and image quality in the assembled configuration. Any aperture also needs a mechanical review.

Make replacement a controlled operation

Define compatible parts, positive seating, clear assembly feedback, instructions and traceability. Establish which lens and serviced variants are covered by the assessment. A replaceable component still needs a controlled replacement process.

THE QUESTION TO BRING TO THE MEETING

Which architecture offers the best verified combination of coverage, useful vision, fit and service?

Fit determines who can use the product.

Prescription support is optical, mechanical and operational work.

Measure prescription demand in the intended workforce or channel. A generic population percentage cannot define the product opportunity. In the US, OSHA describes both prescription-integrated protection and protection worn over prescription lenses without disturbing their proper position. [2]

APPROACH	EVIDENCE TO ESTABLISH
Prescription protective lenses	Supported prescriptions, materials, glazing route, markings, ordering and replacement.
Prescription insert	Complete-system optics, retention, eye clearance, supported variants and fitting process.
Over-glasses arrangement	Stable position with intended frames, visibility, comfort and other protective equipment.

Vertex distance depends on the complete geometry. An insert can move a corrective lens closer to or farther from the eye. Wrap, tilt, pupillary distance, thickness and prescription also matter. Establish the fitting envelope and any compensation with a qualified optical partner.

Assess the insert and its retention under relevant loads. A detachable carrier is not inherently safer. Select representative prescription extremes and configurations with the assessment body.

Run fit trials across head and nose dimensions, movement, hair and required PPE. Record pressure, slip, visual complaints and task performance over representative wearing sessions. Define who the product supports and how the channel identifies exceptions.

THE QUESTION TO BRING TO THE MEETING

Can a real employee order, fit, wear and replace the correct configuration?

Heat sets the terms of sustained use.

Separate electrical safety, wearer comfort and functional continuity.

REQUIREMENT	MEASURE
Safety	Applicable limits, component and battery conditions, foreseeable faults and required protection behavior.
Comfort	Contact-point temperature, pressure, hotspot location and user tolerance over the intended session.
Continuity	Capture, speech, radio and compute behavior as the device warms and thermal controls activate.

Sustainable dissipation depends on area, materials, airflow, solar load, skin contact, geometry and allowable temperature rise. A universal watt limit cannot describe every frame or workload.

Heat moves; it does not disappear

A spreader may reduce a hotspot while warming more surface area. Insulation may reduce heat at a contact point while increasing internal temperature. Validate the full stack, electrical isolation and mechanical behavior in the real geometry.

Measure ambient extremes, sustained and burst loads, radio conditions, battery state and accessories. Log local hotspots as well as averages. Include production variation and ageing in the margin plan.

Define the degraded state

Agree warnings, reduced workloads, session limits and safe shutdown where appropriate. Verify that electronic limiting preserves the intended protective function and leaves the wearer with clear status and a usable recovery path.

THE QUESTION TO BRING TO THE MEETING

What can this configuration sustain in the customer's environment, and what happens when it cannot?

Eight hours. Doing what?

Put the workload next to the battery-life claim.

0.686 Wh	86 mW	26 min
usable energy	average for 8 hours	at a sustained 1.6 W

Illustrative assumptions: 0.95 Wh nominal energy, 85% conversion efficiency and 15% reserve. Usable energy = 0.95 × 0.85 × 0.85 = 0.686 Wh. These hypothetical values explain the method; they are not a product rating.

BUDGET INPUT	WHAT TO CAPTURE
Activity states	Standby, sensing, capture, encode, transfer, playback and processing; measure energy per event and sustained load.
System boundary	Device plus the phone or gateway needed to complete the workflow. Include retries and poor-link conditions.
Replenishment	Charging windows, docks, exchangeable modules, ageing margin and readiness at the start of the shift.

Low-power detection and selected small models may fit locally; larger or sustained workloads may suit a host or cloud. Decide from measured energy, heat, latency, privacy and availability. Local AI and long runtime can coexist when the workload permits.

State continuous capture, standby and mixed use separately. A power interruption needs clear status; passive protection remains a question of physical condition and assessed configuration.

THE QUESTION TO BRING TO THE MEETING

Which activity profile, battery condition and charging workflow support the runtime claim?

A voice feature is an entire exchange.

Who spoke? What was understood? What happened? Could the wearer hear the answer?

STAGE	FAILURE TO LOOK FOR
Wake and detection	Clipped word beginnings or endings; false activation from tools, movement or nearby talkers.
Wearer selection	Confusion between wearer and bystander speech across fit, movement and noise.
Echo and playback	Failures during interruption, profile changes or simultaneous assistant and call audio.
Action and feedback	Wrong actions, misunderstood confirmations, response delay and poor recovery.

Test representative noise spectra, impulses, wind, talker positions, frame movement and protective equipment. Microphone count and a noise-reduction label are inputs, not task-performance evidence.

Pre-roll can preserve speech onset; end-point policy trades completeness against delay. Budget that delay in the whole interaction. Echo cancellation needs an accurate, synchronized playback reference. Choose topology from the implemented audio path and validate every intended route.

Vibration sensing and beamforming may help wearer selection; both depend on fit, movement and processing. Evaluate actual task success, false actions, latency distribution and recovery.

Design feedback around required hearing protection. Use fixtures or controlled recordings where possible, and the workplace safety team for human trials. Voice may need a different feedback channel or task scope when conditions make it unreliable.

THE QUESTION TO BRING TO THE MEETING

Can the wearer complete and confirm the task under the noise and PPE conditions that matter?

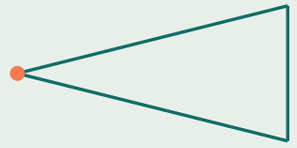
Megapixels describe the sensor. Test the task.

Readable information depends on distance, optics, motion, light and the consequence of a wrong answer.

IDEAL RECTILINEAR GEOMETRY

3,840 px / 839 mm

≈ 4.6 pixels per millimeter



500 mm to target

Schematic; not to scale

Illustrative geometry: for an ideal rectilinear camera and a flat target perpendicular to its optical axis, scene width = 2D × tan(HFOV/2). At D = 500 mm, 80° horizontal FOV and 3,840 horizontal pixels, width is about 839 mm: approximately 4.6 pixels/mm across the ideal target. Real lens distortion and resampling can change local scale. A horizontal 4 mm feature spans about 18 pixels. This is not an OCR pass threshold.

Wider views preserve context; narrower views can place more pixels on a small target at the same distance. Cropping cannot recreate missing detail. Stabilization does not by itself remove blur accumulated during an exposure. Evaluate optics, lighting, exposure, guidance and the complete image pipeline.

Measure the field that the workflow needs

A wrong asset number can look plausible. Measure exact-field accuracy, false acceptance, retries and time to a usable result. Define confirmation or escalation for consequential results. Preserve source evidence where permitted.

Use real target materials, reflectivity, curvature, contamination, distance and motion. Correlate production focus, alignment and window checks with the task acceptance test.

THE QUESTION TO BRING TO THE MEETING

Can the system capture and return the exact required information, and detect when it cannot?

“SDK available” starts the conversation.

A useful interface specification names what the selected platform exposes today.

BOUNDARY	MAKE IT TESTABLE
Microphone	Raw or processed audio, format, sample rate, access route, cloud dependency and allowed concurrency.
Camera	Still capture, recording, preview or streaming; resolution, latency, transport and thermal limits.
Controls and sensors	Events, timestamps, rates, mapping, buffering and behavior during disconnect.
Models and firmware	Permitted deployment targets, memory and compute limits, signing, updates and recovery ownership.
Host and transport	Supported OS versions, permissions, background operation, pairing, reconnection and throughput.

Label each requested capability: **available and evidenced**, **requires development**, **under evaluation** or **unsupported**. Record device revision, firmware and SDK versions, host OS, test conditions and evidence. A platform name alone cannot close integration risk.

Demonstrate the customer's critical workflow on the supported host, including simultaneous capture, audio and transfer where required. Device APIs, application integration, source licensing and ownership are separate deliverables.

Apple's MFi scope depends on the licensed technology used. BLE-only and supported standard-profile accessories without covered technology are generally outside MFi, with specified exceptions. Determine applicability from the design. [6]

THE QUESTION TO BRING TO THE MEETING

Can your application perform the critical workflow on the stated configuration today?

Offline must survive the whole workflow.

Capturing without internet, processing without cloud and operating on an isolated network are different requirements.

LIFECYCLE STEP	DECISION TO CLOSE
Provision and activate	Identity, credentials, initial setup and any online activation or license dependency.
Capture and process	Local, host and cloud boundaries; permissions, encryption, retention and deletion.
Connect and recover	Loss of link, offline queue, retries, duplicate records and incomplete tasks.
Update and support	Firmware and model delivery, recovery, diagnostics, supported lifetime and service ownership.
Reassign and retire	Revocation, secure removal of data, inventory status and end-of-service handling.

Draw the actual audio, image and metadata paths. Name the party that controls each store and the party that responds to an incident. Recording indicators and user permissions are useful controls; they do not independently establish permission for every workplace or jurisdiction.

Validate the intended network restrictions from activation through update and retirement. An offline capture demo leaves licensing, authentication and support dependencies unresolved.

For shared devices, test reassignment and shift turnover. For remote fleets, define diagnostic access, update interruption behavior and an accountable support path. Build graceful failure into the application before deployment.

THE QUESTION TO BRING TO THE MEETING

What still works when the network, host, service or user session becomes unavailable?

The cleaning protocol belongs in the brief.

Qualify the assembled product against the maintenance process people will actually follow.

Cleaning can affect coatings, polymers, adhesives, seals, acoustic paths and markings. Material grade, formulation, residual stress, temperature and dwell time can change the result. Obtain the intended agents and process before promising compatibility.

DEFINE	CHECK AFTER EXPOSURE
Agent and method	Formulation, dilution, wiping, dwell time, rinse, drying and required equipment.
Assembly condition	Stress, material and coating lots, prior ageing and temperature.
Exposure model	Cycles from the use pattern and service interval; inspection and replacement points.
Optics and structure	Haze, transmission, distortion, cracks, retention, bonds and readable markings.
Electronics and audio	Charging, corrosion, ingress paths, acoustic response and wet-state recovery.

Use coupons to screen candidates and assembled parts to qualify the design. Where justified, follow cleaning or ageing with mechanical or ingress evaluation. Keep supplemental sequences distinct from formal assessment requirements.

Evaluate anti-fog behavior after the intended cleaning process. A fresh coating cannot establish performance at a later service interval. Convert test cycles into field-life estimates only with an explicit usage model.

Publish approved cleaning, prohibited methods, inspection and retirement instructions that match the evidence. If a required disinfectant is incompatible, resolve the design or intended use before deployment.

THE QUESTION TO BRING TO THE MEETING

Will the optics, interfaces and instructions still work after the approved maintenance cycle?

The spare lens belongs in the business case.

Compare the cost of an available device and a completed task over the intended service life.

COST LAYER	INCLUDE
Acquire and deploy	Hardware, accessories, integration, training, prescription fitting and initial spares.
Operate	Software or AI usage, connectivity, charging, cleaning, inspection and fleet administration.
Service	Lenses, batteries, labor, shipping, repair turnaround and loaner devices.
Retire and recover	Loss, damage, secure data removal, end-of-life handling and applicable recovery obligations.

A planning equation

Annual service cost = service events/year × delivered cost/event + downtime cost + support overhead.

Count the part, labor, logistics, inspection and any required requalification in delivered service cost. Obtain event frequency from the intended environment. Keep software and fleet operating cost separate to avoid double counting.

User-replaceable lenses can preserve working electronics and reduce downtime. Factory repair or replaceable front modules can also be viable. Compare alternatives on the same usage and response-time assumptions.

Control compatible parts, traceability, replacement instructions and retirement criteria. Treat changes to safety parts through the same change process as the original configuration. Model benefit from pilot evidence; do not book unverified productivity as guaranteed savings.

THE QUESTION TO BRING TO THE MEETING

When a lens is scratched or a battery ages, what is the delivered cost and time to restore service?

Schedule the evidence that unlocks shipment.

Connect the calendar to representative samples, external dependencies and controlled changes.

WORKSTREAM	EARLY AGREEMENT
Protection	Hazards, markets, variants, sample matrix, documentation and assessment route. [1-4]
Electrical and radio	Applicable electrical safety, EMC, radio and head-worn RF-exposure evaluations.
Battery and transport	Cell versus pack evidence, finished configuration and shipping documentation. [7]
Wireless and platforms	Qualification owner, permitted design reuse and licensed-technology scope. [6,8]

Formal assessment may overlap design validation when the body accepts representative samples and the configuration is stable enough. Allocate responsibility for subsequent changes, extra samples and retesting. Neither automatic full retest nor automatic exemption is a sound change policy.

Separate functional demonstration, validation, required assessment, production readiness and launch dates. Obtain actual tooling queues, long-lead commitments, laboratory availability and supplier shutdown and recovery dates.

An engineering-sample label does not itself authorize shipment or workplace use. Controlled evaluation needs an appropriate setting and defined restrictions. An unassessed prototype must not substitute for required eye protection.

Before release, connect the production revision to the applicable reports, declarations or certificates, labels, instructions, quality release and service information.

THE QUESTION TO BRING TO THE MEETING

Which evidence is on the critical path, and which configuration will it actually cover?

Each gate should earn the next investment.

A stage name creates little value until its acceptance evidence is defined.

GATE	DECISION EVIDENCE
Definition	Primary task, baseline, user group, markets, target economics, claims and initial risks.
Architecture	Interfaces, physical budgets, optics and fit, data flow, supplier feasibility and validation plan.
EVT · Engineering validation	Measured evidence that the engineering concept works; failures, root causes and corrective action.
DVT · Design validation	Representative builds assessed against approved requirements, with configuration and issue control.
PVT · Production validation	Process flow, work instructions, control plan, test correlation, yield, traceability and supplier readiness.
Release and monitoring	Applicable market-access evidence, approved claims, service readiness and field-response ownership.

Use requirements, risk analysis and validation plans together. FMEA and related methods can improve discipline; their use does not assert certification to an unrelated quality framework.

Production checks should detect the failures the process can create: focus, alignment, acoustic response, charging, assembly integrity and firmware identity as applicable. Demonstrate measurement repeatability and correlation with the design requirement.

Control substitutions, rework, nonconforming units and returns by configuration. Assess changes by consequence, including firmware, radio, thermal behavior and claimed performance. Resolve critical safety issues before the relevant release.

THE QUESTION TO BRING TO THE MEETING

What evidence closes this gate, and who owns every unresolved release-blocking issue?

Reuse has three assets: technology, evidence, rights.

Price the verified starting point and the remaining work.

STARTING POINT	POSSIBLE REUSE	WORK TO ESTABLISH
New architecture	Components, methods or tools.	System feasibility, bring-up, validation and production route.
Existing electronics; new protective assembly	Compatible electronics, firmware and interfaces.	New physical assembly, fit, thermals, acoustics, protection and process.
Derivative of an assessed product	Permitted configuration evidence and technical assets.	Change scope, variant coverage and additional tests or documentation.

Platform maturity can reduce risk, time and cost. The benefit depends on what remains usable, supported and permitted. A new frame does not automatically inherit protection evidence; a reusable report is not the same as automatic approval.

Build a scope-normalized quotation

Separate development NRE, tooling and fixtures, external assessment and licensed technology, validation builds, recurring production cost and contingency. Name the disciplines, iterations, samples, software deliverables and acceptance included in each line.

Document supplier continuity, tool access, evidence applicability and permissions. Estimate schedule from dependencies and actual resources. One program's totals cannot set a universal price or timeline for a different architecture.

THE QUESTION TO BRING TO THE MEETING

Which assets remain unchanged, which evidence still applies and which rights permit the intended use?

Access, ownership and support are separate.

Make the deliverables and their usable rights visible before the project depends on them.

AGREEMENT TOPIC	QUESTION TO RESOLVE
Background and new IP	What exists already, what is created, who owns it and which licenses permit the intended use?
SDK and source	Which APIs, binaries, files, build tools and documentation are included; what may be modified or shared?
Third-party technology	Which component-vendor, open-source, commercial and platform permissions apply independently?
Tools and continuity	Who owns, maintains and operates tooling, and what manufacturing, transfer and support rights exist?
Acceptance and changes	Which observable deliverables trigger review and payment; who carries iteration, retest and scope changes?
Publicity and modern tools	What may be published or processed in cloud or AI development systems, under what agreed conditions?

Define the cost base and whether a commercial percentage is margin or markup. Agree treatment of yield, warranty, supplier movements and currency where relevant. Compare proposals on common scope.

Permission to access a repository does not establish ownership, redistribution rights or permission to process protected material elsewhere. A component's open-source license does not grant rights to unrelated designs or marks.

Public capability statements need support and permitted disclosure. Distinctive parameters, calendars or commercial terms can identify a project even when names are removed. Agree publicity and source-processing rules in the actual engagement.

THE QUESTION TO BRING TO THE MEETING

Can each party use, maintain and commercialize the deliverables in the way the business plan assumes?

Measure useful work and the cost of failure.

Set acceptance criteria before the trial, then report the conditions and the denominator.

MEASURE	PRACTICAL DEFINITION
Task success	Correctly completed tasks / all attempted tasks; include retries, abandonment and recovery.
Time to completion	Baseline and trial distribution; include setup, confirmations and correction of failed attempts.
False acceptance	Wrong outputs accepted as correct / defined evaluated cases; state the denominator and review method.
Wearer acceptance	Completed wearing sessions, removal reasons, discomfort, fit exceptions and interaction issues.
Operational availability	Ready-for-use periods / planned periods; show charging, repair and software outage separately.
Economics	Measured operating and service costs against attributable task benefit, using stated assumptions.

Represent the users and environments intended for deployment. Record build and software versions, training, sample size, task mix and excluded conditions. Small pilots leave uncertainty; report the limits instead of treating an observed percentage as universal performance.

For AI outputs, preserve a route to confirmation, correction or escalation when consequences warrant it. Trace answers to captured evidence or approved instructions where appropriate. Plan behavior for uncertainty, stale content and unavailable services.

Agree who decides whether to expand, redesign or stop. A promising result becomes more useful when the next investment and remaining evidence are explicit.

THE QUESTION TO BRING TO THE MEETING

Would the same trial design expose an unsuccessful product as clearly as a successful one?

Fifteen questions before commitment.

For each answer, record: owner · decision basis · missing evidence · due date.

01	Which task improves, and how will success be measured against today's workflow?
02	Who wears it, for how long, and with which prescription needs and other equipment?
03	Which hazards, markets, configurations and public claims define the protection brief?
04	Which impact, ingress, perspiration, cleaning and ageing exposures must it withstand?
05	What lens, frame and prescription architecture fits the intended user envelope?
06	Which thermal and battery behavior is acceptable at the intended workload and ambient?
07	What speech criteria apply under the actual noise, motion and hearing protection?
08	Which targets and error consequences define capture and OCR acceptance?
09	Which functions run on the device, host or cloud; what happens when one is unavailable?
10	Which interfaces, host versions, data policies and fleet functions must be demonstrated?
11	How are lenses, batteries and service parts cleaned, replaced, inspected and traced?
12	Which technology, evidence and rights can be reused, with what support continuity?
13	What configuration is tested, who controls changes and what evidence authorizes release?
14	What are the full development, operating and service costs, dependencies and reserves?
15	Who owns deliverables, source rights, tooling, acceptance, support and publicity decisions?

THE QUESTION TO BRING TO THE MEETING

Leave the review with a decision register that both the buyer and engineering team can use.

Bring the workflow. Build the evidence. Define the product.

AI and AR eyewear development for brands and solution providers.

SmartXY (Shenzhen Xingyi Intelligent Technology Co., Ltd.) provides ODM and JDM engineering for smart eyewear. Scope can bring together product definition, system architecture, mechanical and electrical design, firmware, acoustics, thermal work, prototypes, manufacturing preparation and coordination of external testing.

START HERE	WHAT WE ESTABLISH TOGETHER
1 · Share the brief	Target user, primary task, markets, protection needs, software environment, economics and launch priorities.
2 · Evaluate the starting point	Platform fit, capability gaps, architecture trade-offs, key risks and the verification route.
3 · Define the engagement	Deliverables, responsibilities, acceptance evidence, commercial assumptions and dependencies.

The first useful outcome is a reviewable product brief, feasibility scope and development plan. Platform selection, capability availability, software deliverables and manufacturing arrangements are confirmed for each project.

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Every performance claim needs a workload, a configuration and a measurement.

Sources that support the decision.

Public-source status checked 8 September 2026.

[1] ISEA · ANSI/ISEA Z87.1-2025 publication
Confirms publication and scope; it does not establish certification of a named product.

[2] OSHA · 29 CFR 1910.133
Workplace eye and face protection, including prescription arrangements. New ANSI publication and regulatory adoption are distinct.

[3] CSA Group · Z94.3 edition collection
Official collection lists Z94.3:26 as active and Z94.3:20 (R2025) as withdrawn. Confirm the edition applicable to the project.

[4] BSI · Understanding EN ISO 16321
Discusses EN ISO 16321-1:2022+A1:2025 and the EN 166 transition. Confirm the exact current harmonized reference and applicable route.

[5] IEC · Ingress Protection ratings
Public overview of enclosure ingress classifications under IEC 60529.

[6] Apple · MFi Program FAQs
Explains licensed technology and exclusions, with exceptions, including BLE and supported standard profiles.

[7] US PHMSA · Lithium Battery Test Summaries
Explains design-test and summary requirements; check the actual cell, pack and shipping configuration.

[8] Bluetooth SIG · Qualify your product
Consult for qualification responsibility, process and current requirements.

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